

Trend Following Technical Report

Overview

- The Trend Following strategy is based on fact that market up-down movements are not just random.
- But they are driven by major big players and emotional habits of other participants.
- So we will trade by finding momentum, its strength and confirming it.

Common Used Terms

Close : Closing Price of a Equity on a particular day

EMA : Exponential Moving Average which is basically exponential mean of last n days, where n is generally called as window size.

where we give more weightage to just previous prices.(Formula - $EMA_t = Price_t * Multiplier + EMA_{(t-1)} * (1 - Multiplier)$, where $Multiplier = 2/(n+1)$).

True Range : It Basically measures the true market volatility.

Indicators

Name	Formula	Interpretation
<i>MACD</i>	EMA12-EMA26	MACD>Signal Line → bullish MACD<Signal Line → bearish
<i>True Range</i>	$\text{MAX}(H-L, H-C_{(t-1)}, L-C_{(t-1)})$	Used for DI calculation
<i>+DM</i>	$H - H_{(t-1)}$ if it is greater than $L - L_{(t-1)}$ and 0	for DI calculation
<i>-DM</i>	$L - L_{(t-1)}$ if it is greater than $H - H_{(t-1)}$ and 0	
<i>+DI</i>	smoothed +DM/smoothed true range	+DI>-DI → bullish
<i>-DI</i>	smoothed -DM/smoothed true range	+DI<-DI → bearish
<i>ADX</i>	$100 * (+DI - -DI) / (+DI + -DI)$	Trade when ADX>25
<i>Signal Line</i>	EMA9 of MACD	Used for MACD indicator

My Strategy By Indicator

ADX :

It is indicator which used to determine the strength of the trade, it does not tells about the bullish or bearish trend of market, for that we use another indicators.

It varies from 0 to 100. and above 25 is considered as strong strength so we trade only when $adx > 25$.

MACD:

It is basically Moving Average Convergence Divergence , It is used for determining the direction of the trend and momentum.

It is calculated by difference of two exponential moving averages(EMA12-EMA26).

we also calculate a signal line (EMA9 generally) which helps macd to determine direction.

If it crosses above the signal line there is a bullish signal and if below then bearish signal.

+DI AND -DI:

These are directional indicator helps in finding the trend direction generally used for coonfirming the trend direction.

if +DI>-DI we can say there is bullish signal and if +DI<-DI then bearish signal.

Drawbacks of these Strategy:

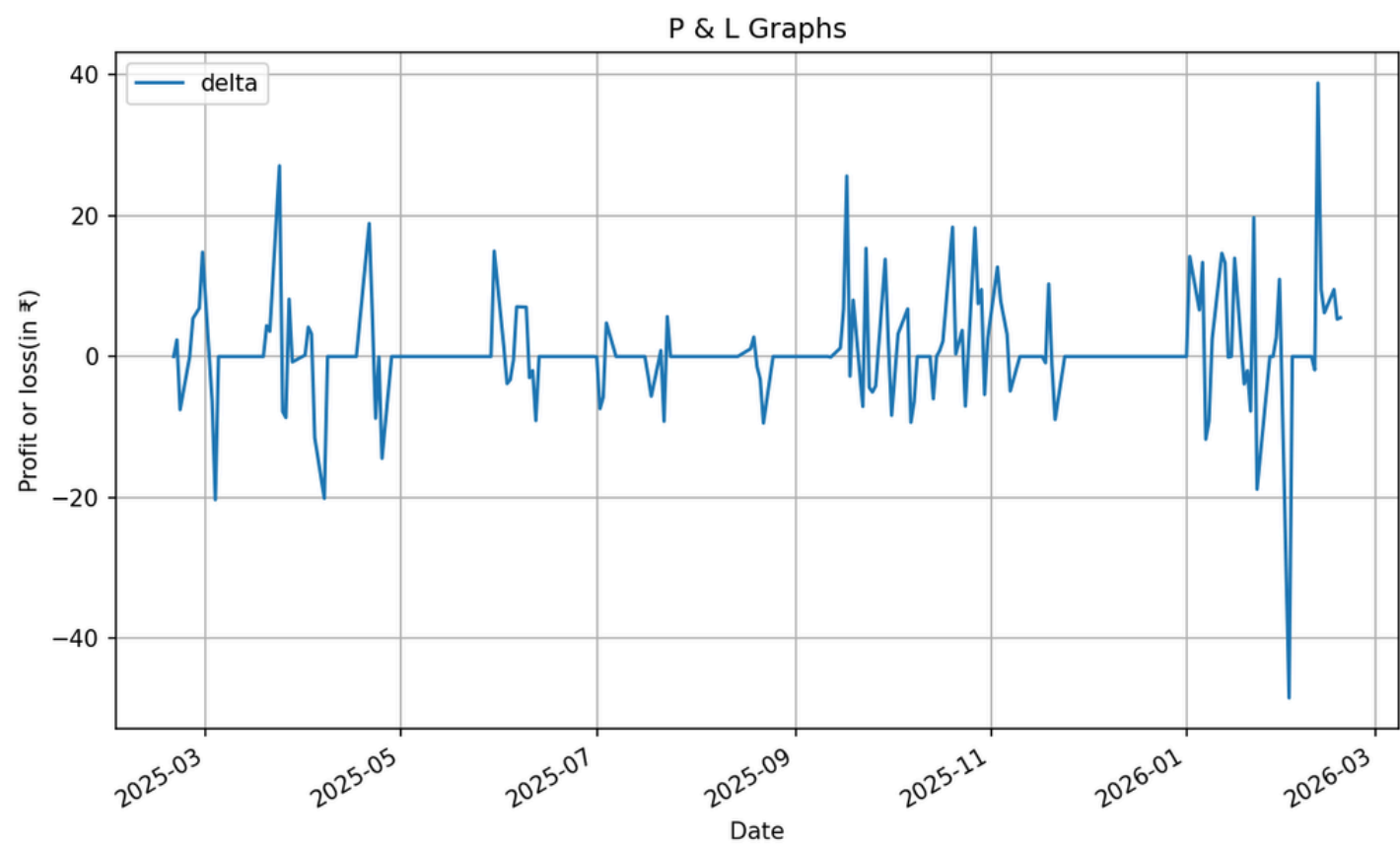
- In range bounds market it doesn't perform well
- It works best in high liquidity market.

Results

Equity	Sharpe	Annual Return	Max Drawdown	Average Profit(in INR)
<i>SBI</i>	<i>1.275736</i>	<i>16.7%</i>	<i>-0.066307</i>	<i>8.608351</i>
<i>INFOSYS</i>	<i>0.798630</i>	<i>11.6%</i>	<i>-0.113946</i>	<i>20.785097</i>

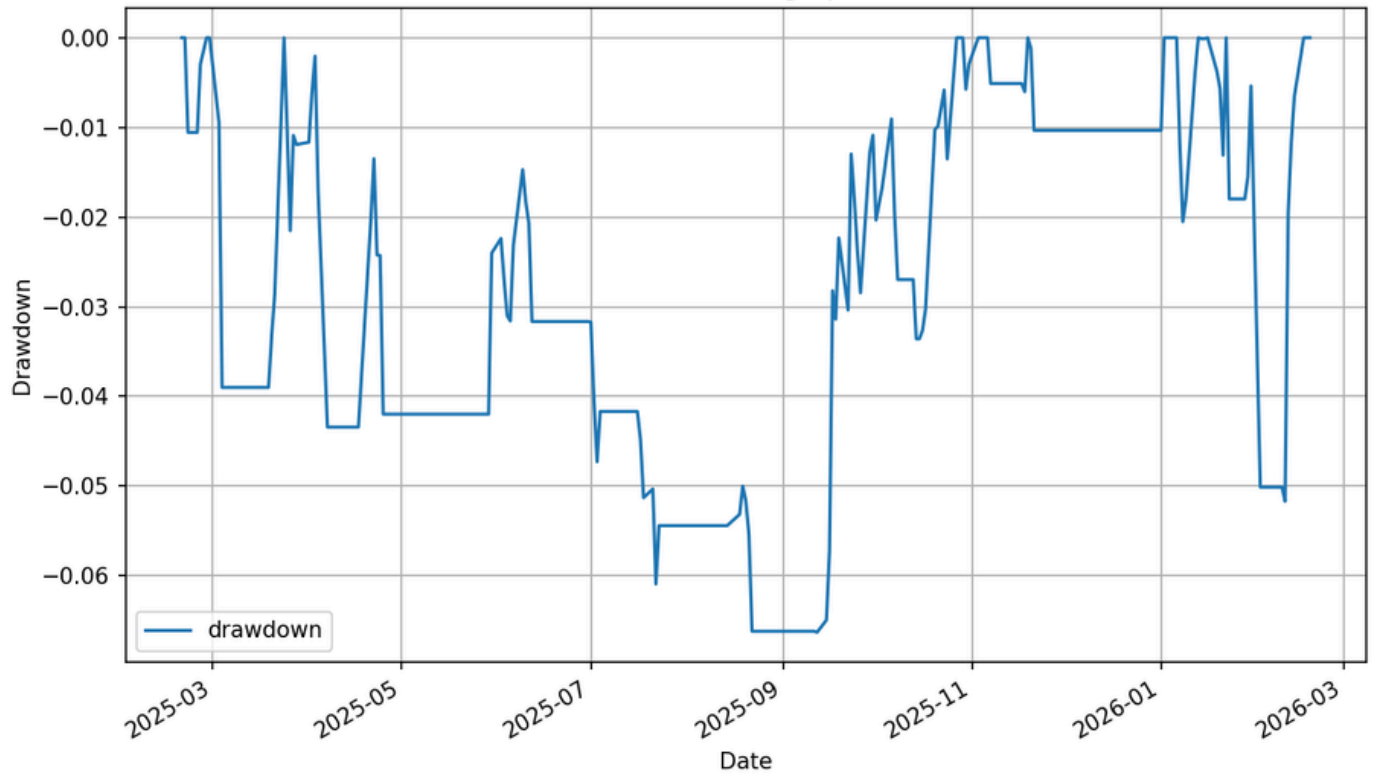
Graphs

P & L Graph for SBI:



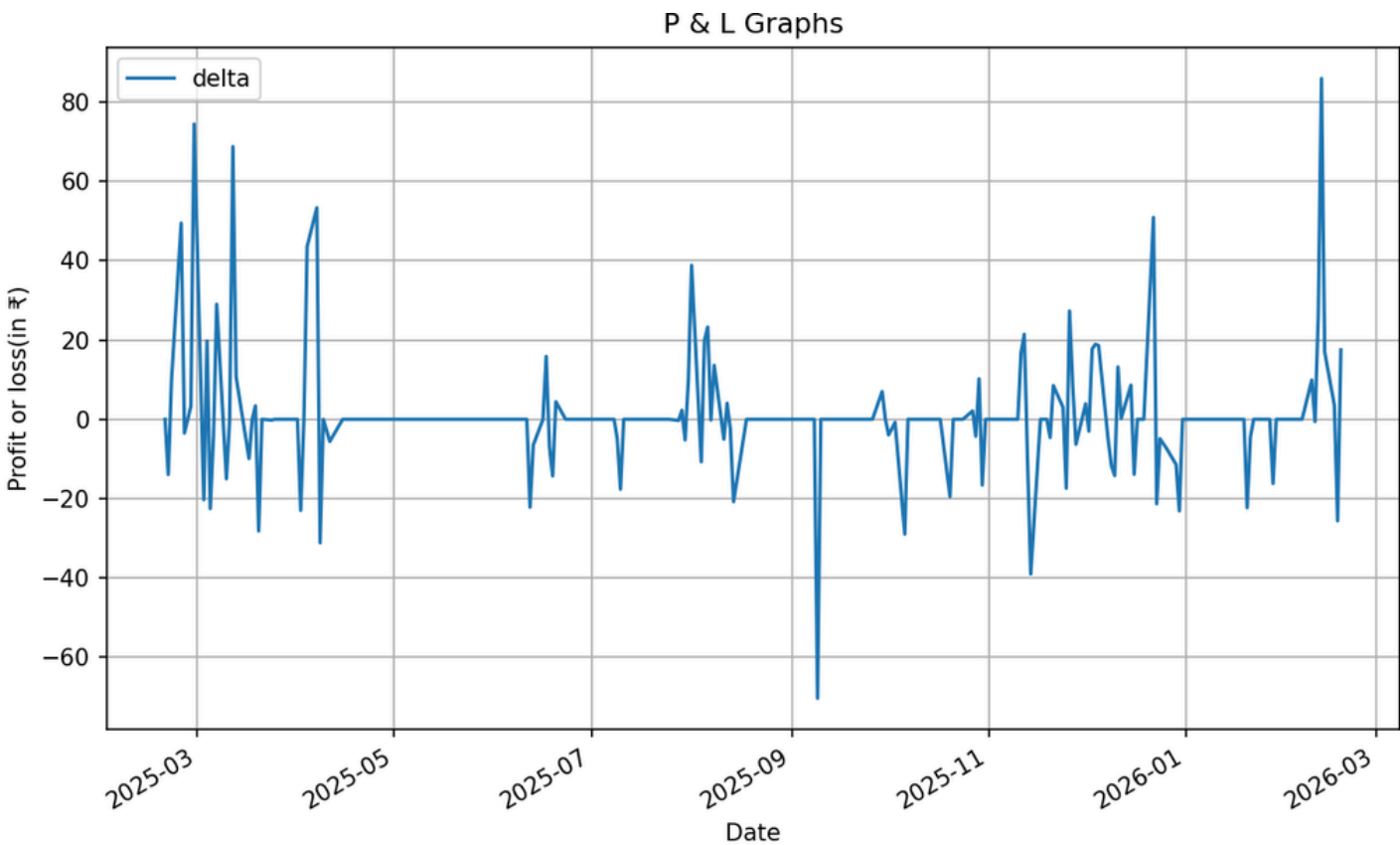
Drawdown Graph:

Drawdown graph



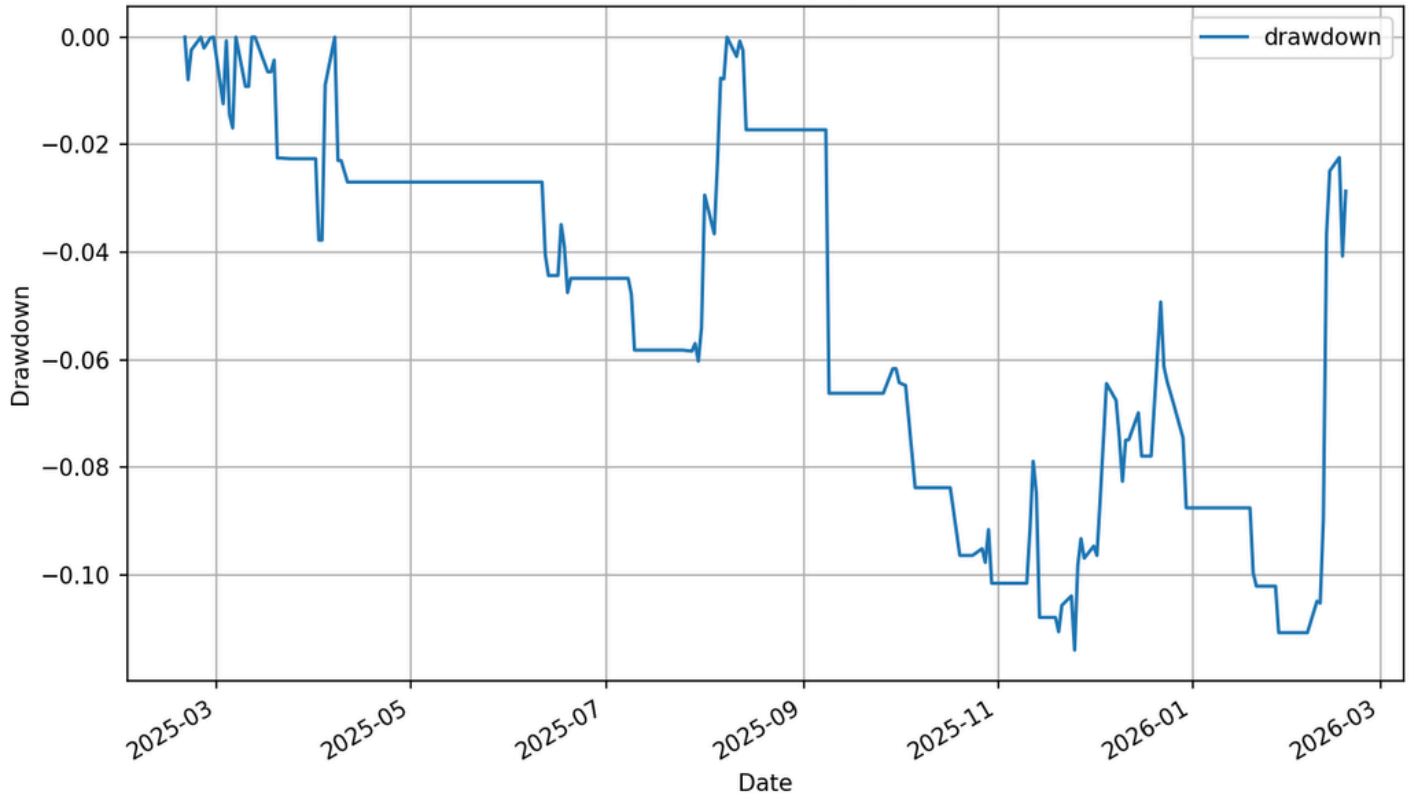
Graphs

P & L Graph for INFOSYS:



Drawdown Graph:

Drawdown graph



Interpretation from results

- That Strategy behaves good in high liquid stocks and stocks which volatility is rising is better for it(like- SBI)
- Since we are using too many constraints therefore the winning rate is not such impressive.
- Trend Following strategy is generally good for all but it shouldn't be a range bound stock.

Resources:

- [🌐 Average Directional Index \(ADX\): Definition and Formula](#)
- [What Is MACD?](#)
- [How to Use ADX \(Average Directional Index\) in Forex - Babypips.com](#)
- Learned mostly from google sites and chatgpt.